105 Gases

1.5.1 Pressure

Pressure is normal force per unit area. The same force concentrated on a smaller area produces greater pressure. Use the component perpendicular to the surface and the actual area over which it acts, not an arbitrary force component. One pascal is one newton of normal force per square metre.

$$p = \frac{F_{\perp}}{A}$$

p: pressure (Pa); F⊥: force normal to the surface (N); A: area (m²).

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1.5.2 · 1.5.3 Boyle's law

For a fixed mass of gas at constant temperature, absolute pressure is inversely proportional to volume, so the pressure-volume product is unchanged between two states. Halving the volume doubles the

absolute pressure. Use absolute rather than gauge pressure; the two-state relation cannot be applied unchanged if gas leaks or temperature varies.

$$p \propto \frac{1}{V}$$

$$pV = \text{constant}$$

$$p_1V_1 = p_2V_2$$

Boyle's law: fixed mass of gas at constant temperature. p: absolute pressure; V: volume.

Fixed mass of gas at constant temperature; p: absolute pressure; V: volume. Use consistent units in both states.

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1.5.4 Celsius and kelvin temperatures

The kelvin scale starts at absolute zero. Add 273.15 to the Celsius reading to obtain the kelvin value; the textbook commonly rounds this to 273. A rise of 1 K equals a rise of 1 °C, but ratios of the readings differ. Gas laws and absolute-temperature proportionalities require kelvin.

$$\frac{T}{\mathrm{K}} = \frac{\theta}{^{\circ}\mathrm{C}} + 273.15 \approx \frac{\theta}{^{\circ}\mathrm{C}} + 273$$

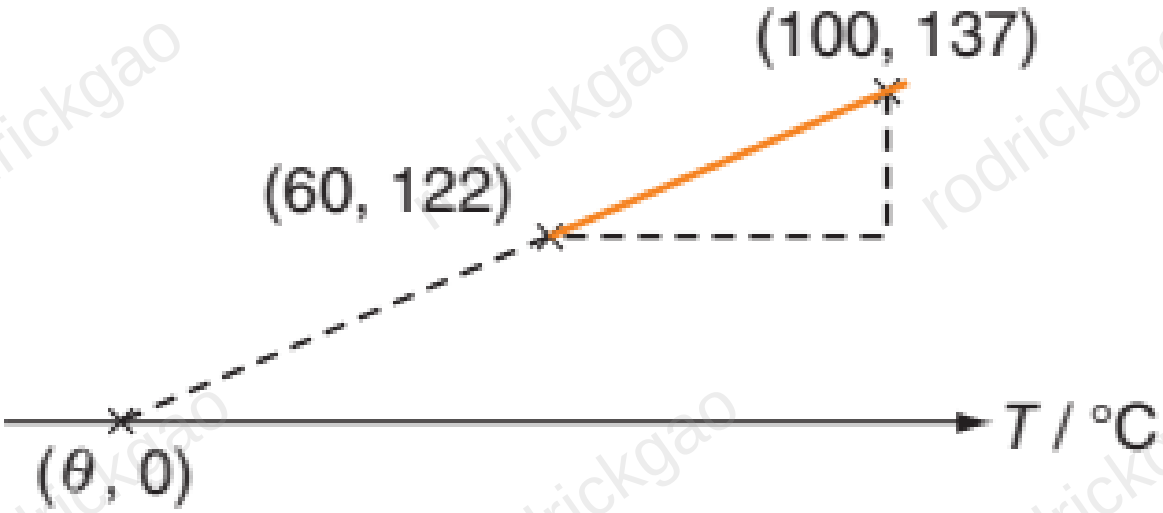
T: thermodynamic temperature; θ: Celsius temperature. The textbook uses the rounded offset 273; 273.15 is exact.

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1.5.5 Extrapolating a straight-line graph

At fixed gas amount and volume, pressure approximately follows $p = m\theta + b$ against Celsius temperature. Two points give $m = \frac{p_2 - p_1}{\theta_2 - \theta_1}$; extending to zero pressure gives the estimated absolute zero at $\theta_0 = -\frac{b}{m}$. This is extrapolation beyond measurements, not actually cooling gas to zero pressure. The textbook data give −265 °C rather than the accepted approximately −273 °C; use the data and discuss experimental error.

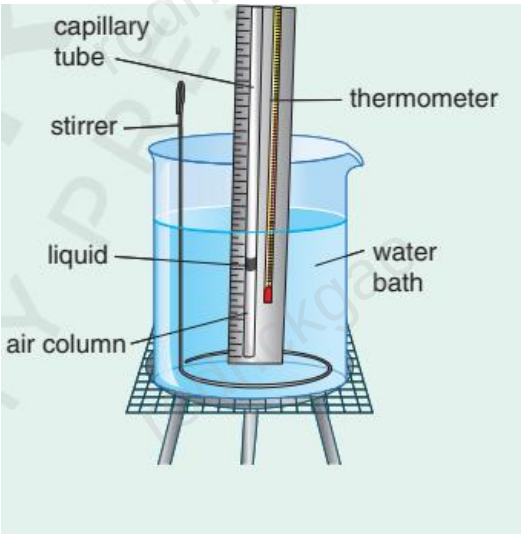


Extrapolate the pressure-Celsius-temperature line to zero pressure.

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1.5.6 Uniform temperature and steady readings

Stop heating and stir the water bath thoroughly before reading so its temperature is more uniform, then wait for a steady reading. Stirring addresses spatial temperature differences; waiting allows the sample and bath to approach thermal equilibrium. The textbook also warns against letting the thermometer touch the beaker bottom.



The experimental arrangement includes a water bath, thermometer and gas sample.

Textbook sources: 105 · PDF 12, 15

1.5.7 · 1.5.8 Pressure and temperature at constant volume

For a fixed mass of gas at constant volume, absolute pressure is proportional to kelvin temperature. Heating increases the mean momentum transferred to the container walls in collisions. Keep pressure units consistent, convert Celsius readings to kelvin and verify that neither the container volume nor the amount of gas has changed.

$$p \propto T$$

$$\frac{p}{T} = \text{constant}$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

Fixed mass and fixed volume; p: absolute pressure; T: temperature in kelvin.

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1.5.9 · 1.5.10 Volume and temperature at constant pressure

For a fixed mass of gas at constant pressure, volume is proportional to kelvin temperature. Heating allows the gas to expand while maintaining its pressure. Compare volume divided by absolute temperature in the two states. Do not use Celsius readings directly or apply the relation when gas leaks or pressure changes appreciably.

$$V \propto T$$

$$\frac{V}{T} = \text{constant}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Fixed mass and constant pressure; V: volume; T: temperature in kelvin.

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1.5.11 · 1.5.12 The combined gas law

For a fixed amount of ideal gas, pressure multiplied by volume and divided by kelvin temperature remains constant. This relates changes in all three quantities. Before comparing states, check that no gas enters or escapes, pressure is absolute, and pressure and volume units are consistent between states. Temperature must be in kelvin.

$$pV \propto T$$

$$\frac{pV}{T} = \text{constant}$$

$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$

Combined gas law for a fixed mass: p is absolute pressure, V is volume, and T is in kelvin. Neither pressure nor volume must be held constant.

Fixed amount of ideal gas; p: absolute pressure; V: volume; T: kelvin temperature.

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1.5.13 The ideal gas equation

The ideal gas equation relates absolute pressure, volume, amount of substance and kelvin temperature. It can also describe separate equilibrium states when the amount of gas changes. With the SI gas constant, use Pa, m³ and mol. The number of molecules is not the number of moles.

$$pV = nRT$$

p: pressure (Pa); V: volume (m³); n: amount (mol); T: temperature (K); R = 8.31 J mol⁻¹ K⁻¹. Ideal gas.

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1.5.14 The microscopic origin of gas pressure

Gas molecules move randomly and change momentum when they collide with container walls, exerting forces on those walls. The mean force from many collisions divided by wall area gives the pressure. Greater molecular number density or greater mean speed can raise pressure; gas pressure is not caused solely by the gas's weight.

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1.5.15 Assumptions of the kinetic theory of gases

The model treats a gas as many identical molecules in continuous random motion. Molecular volume and collision duration are negligible on the relevant scales; intermolecular forces are ignored except

during collisions. Collisions between molecules and with walls are perfectly elastic. These assumptions simplify the average collision effects but do not hold exactly for every real gas under all conditions.

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1.5.16 Gas pressure and mean square speed

Kinetic theory relates gas pressure to the number of molecules, the mass of one molecule and mean square molecular speed. The bar means square each speed before averaging, not square the average speed. The model assumes isotropic random motion. Use the number of molecules, not the number of moles, in the molecular form.

$$pV = \frac{1}{3}Nm\bar{c^2}$$

p: pressure (Pa); V: volume (m³); N: number of molecules; m: mass of ONE molecule (kg). The overbar over c² means the mean square speed (m² s⁻²), not the square of the mean speed. Ideal gas with isotropic random motion.

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1.5.17 · 1.5.18 Absolute temperature and mean molecular kinetic energy

The mean translational kinetic energy per ideal-gas molecule depends only on kelvin temperature. Different gases at the same temperature therefore have equal mean translational kinetic energies, although heavier molecules have smaller root mean square speeds. This is an average per molecule, not the total kinetic energy of the whole sample.

$$\langle K \rangle = \frac{1}{2} m \bar{c^2} = \frac{3RT}{2N_A}$$

$$T \propto \langle K \rangle$$

⟨K⟩: mean translational kinetic energy per molecule (J); m: one molecule's mass (kg); mean square speed: $m^2\ s^{-2}$; T: K; R = 8.31 J mol^{-1} K^{-1}; N_A = 6.02 × 10^{23} mol^{-1}.

For an ideal gas, T is absolute temperature (K), and ⟨K⟩ is mean translational kinetic energy per molecule (J).

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